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BOOK REVIEW

The psychology of science denialism

Gale M. Sinatra and Barbara K. Hofer: Science denial: why it happens and what to do about it. Oxford: Oxford University Press, 2021, 208pp, £22.99 HB

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Never before has the entire world been more aware of the dangers of science denialism than with the COVID-19 pandemic that we just went through. The anti-vaccine movement and other science denial groups have caused really pressing problems for public health in many countries that, paradoxically, stand out as leading scientific hubs for scientific research.

This calls for urgent action to counter science denialism and promote a scientifically informed civil society. A scientific mindset, with its emphasis on evidence-based reasoning and rational evaluation, as well as tolerance and open-mindedness, can help build better democratic societies (Koertge 2005). For these tasks of promoting science, the discipline of psychology can be of great help. Psychological science deals with the cognitive, affective and motivational mechanisms that lie behind our behavior and thought processes, and how these are influenced by social and developmental factors. This includes, of course, the cognitive, affective and motivational mechanisms that underlie our anti-science thoughts and beliefs. Although psychology alone may not be enough to counteract science denialism, effective advocacy and promotion of a scientific attitude among the general population require taking into account the data, hypotheses and constructs put forward by psychologists. For this task, the book at hand, *Science denial: Why it happens and what to do about it,* by educational psychologists Gale M. Sinatra and Barbara K. Hofer, can be of great help.

Sinatra and Hofer's book constitutes an extensive literature review on the psychology of science denial that can help lay readers to digest what psychologists have learned about this topic during the last decades of research. Though perhaps too America-centered, the book can be of great use not only to psychologists themselves, but also to empirically minded philosophers, science communicators and

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educators alike. Of particular interest are the series of practical recommendations that Sinatra and Hofer put forward to promote science understanding and acceptance in very specific settings, such as, for instance, classrooms or newspaper newsrooms.

The book is structured in two sections. The first section comprises three chapters and presents the problem of science denial as well as the potential role of science education in mitigating it. The second section tries to provide five non-exclusive explanations for science denial, based on five different psychological constructs well known to psychologists: cognitive bias, epistemic cognition, motivation, emotion and attitudes. Finally, the book ends with a chapter summarizing how to prevent science denial.

Chapter 1 introduces the problem of science denial, offering justifications for why it is a really pressing problem. The main point of the chapter is that science denial is not only worrying because it contributes to undermine something valuable—science and scientific inquiry—but also, as suggested before, because of its bad consequences. Science denial has been common since at least the Scientific Revolution, but recent changes in our society have caused it take on new forms and ways of manifesting itself. That is why it is important to know the psychology of science denial.

Chapter 2 focuses on how to deal with science claims online. Most people today search for information on scientific topics on the Internet. However, several problems stand in the way of getting online information right. Among these, the most pervasive are the unreliable and unconscious ways we often use to assess the credibility of Internet sources, our tendency to favor personal anecdotes over statistical data or the way search algorithms are designed, which tend to favor filtering bubbles that reinforce our own prejudices. Taken together, all of these factors can lead to science denial.

Chapter 3 discusses the role of science education in combating science denialism. Americans are not particularly good at science. The poor preparation of teachers and the relatively minor role of scientific knowledge in the curriculum are partly responsible for this situation. However, contrary to what it may appear at first glance, the authors argue that increasing scientific knowledge is unlikely to solve the problem. The solution may lie, instead, not through teaching more scientific facts, but by teaching more scientific practice. Science textbooks often present an utterly simplified and misguided view of how scientists conduct research; the authors argue that a more engaging, active and practice-centered science education would help students to better understand what science is and how to relate scientific thinking to everyday questions.

The second section of the book begins with Chapter 4, which explores how cognitive biases can lead us into science denial. Most of us consider ourselves much more rational than we actually are. But the fact is that several cognitive processes stand in the way of understanding scientific claims. For instance, humans tend to rely too heavily on what psychologist Daniel Kahneman has called the "System 1" of thought. System 1 can make us see the world as much simpler than it really is, and it can lead us to focus only on evidence that confirms our preexisting beliefs. On top of that, our intuitive theories about how things work, the illusion of understanding, or our common tendency to misjudge the breadth of our knowledge on a certain topic can also play a role in breeding science denial.

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Chapter 5 deals with the importance of epistemic cognition. People not only have beliefs about particular scientific topics, but also beliefs about the nature of scientific knowledge itself. These beliefs are often referred to as "epistemic cognition." Certain epistemic beliefs can hinder the understanding and acceptance of science, while others can facilitate it. For instance, while absolutist and multiplist (i.e., relativist) views of knowledge tend to run counter to a scientific mindset and they can eventually breed science denial, evaluativism, or the view that evidence should play a crucial role in evaluating the validity of empirical claims, can foster it. Psychological evidence suggests that formal education and societal factors can affect the development of one or another type of epistemic cognition. For example, "false equivalence" debates in science journalism can contribute to relativistic views on knowledge. A series of common misconceptions about the nature of science can also stand in the way of developing evaluativist epistemic cognition. These include, for instance, wrong ideas about what scientific theories are or what role uncertainty plays in science.

Chapter 6 analyzes the role of motivations in science denialism. Motives other than purely epistemic ones, i.e., economic, social, political, etc., can influence the way we tend to interpret evidence, eventually leading to a rejection of scientific claims. For instance, evidence has shown that political conservatism is associated with skepticism about certain scientific topics, although motivational biases affect both sides of the political spectrum. The authors recognize social identity as another powerful force driving motivated reasoning about science, preventing people from adopting a scientific standpoint if it collides with their group's worldview. For this reason, showing potential science deniers that their identities, especially religious ones, need not conflict with the acceptance of scientific evidence is crucial to promoting trust in science.

Chapter 7 examines the literature on the influence of emotions and attitudes in our understanding and acceptance of science. Positive emotions tend to reinforce science knowledge because they make people more likely to engage in active learning strategies. The problem, of course, is that certain topics tend to elicit negative emotions more often, e.g., evolution and climate change. That is why efforts should be made to promote positive emotions when teaching controversial topics, for instance, highlighting how seemingly small personal actions can help combat climate change. An analogous case can be made for attitudes: positive attitudes toward science tend to promote positive feedback loops that favor learning, while negative attitudes tend to breed mistrust and rejection of science.

Sinatra and Hofer's book ends with Chapter 8, which summarizes everything that can be done to prevent and fight science denial at different levels, from individual actions to policy making. I will mention some examples. On an individual level, we can try to develop a scientific attitude and an appreciation for truth, monitor our own cognitive biases, advance our algorithmic literacy and vote for science-friendly parties. At the science education level, teachers can take into account the previous beliefs, attitudes and identity of their students and try to engage them in scientific practice as applied to real problems. At the level of science journalism, science communicators can push media to pay more attention to science, not allow science deniers to have an equal footing in debates and emphasize what methods scientists

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use to reach their conclusions. Finally, policymakers can support science education at all levels and use science to design their policies.

Science denialism concerns all of us who consider ourselves friends of science, whether we are scientists, philosophers, educators, journalists or policymakers. Science denialism is a complex phenomenon that requires multifaceted measures to counter it. These measures should certainly take into account what psychologists have learned about the cognitive, affective and motivational factors involved in rejecting science. That is why *Science denial: Why it happens and what to do about it* is an important book.

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Reference

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